ENGINEERING MATERIAL SELECTION GUIDE

JLCPCB / JLC3DP 3D Printing Materials

Strength, cost, manufacturability, and component recommendations for a quadruped robot

Decision in one sentence

Use MJF PA12-HP or SLS PA12-family nylon for most load-bearing body and leg structures; reserve PA12-CF for stiffness-dominated parts that can tolerate FDM anisotropy, PA11 for impact-dominated parts, TPU/PEBA for feet, and printed metal only for compact interfaces where polymer is demonstrably insufficient.

Prepared for

Quadruped Robot Project

Research date: 26 August 2026 | Catalog availability and prices can change

Scope: Publicly listed JLC3DP/JLCPCB additive-manufacturing materials and service specifications. The document distinguishes published values from engineering screening judgments. It is not a substitute for coupon testing, FEA, fatigue testing, or a supplier-certified material lot report.

1. Executive Recommendation

Recommended material stack

A mixed-material robot is the technically strongest and usually the cheapest solution. Do not force one material to serve as structure, bearing interface, heat shield, cable protector, and foot tread.

Component	Primary choice	Why	Alternate / boundary
Main body / chassis shells	MJF PA12-HP	Best overall balance of toughness, fatigue resistance, dimensional repeatability, complex geometry, and low mass.	SLS 3301PA or 3201PA-F when quote cost or size is better.
Thigh and shank link shells	MJF PA12-HP or SLS 3301PA	Tough powder-bed nylon tolerates impacts and has less directional weakness than FDM.	PA12-CF only when stiffness is the dominant problem and orientation is controlled.
Impact-prone clips / compliant brackets	MJF PA11-HP	JLC3DP publishes 50% elongation at break and equal XY/Z tensile strength.	PA12-HP for lower cost or better dimensional familiarity.
Actuator brackets / bearing carriers	TC4 SLM for weight-critical; 316L SLM for cost-tolerant mass	High strength and better creep resistance at concentrated loads.	Powder nylon with metal sleeves only after load testing.
Gearbox housings / precision bores	Metal interface or PA12-CF shell plus machined metal inserts	Printed polymer alone is vulnerable to creep, bore growth, and bearing-seat damage.	MJF PA12 with large wall sections for low-load prototypes.
Foot pads / bumpers / strain relief	TPU 95A	Abrasion resistance, replaceability, and compliance.	PEBA 85A when elastic energy return matters more than damping.
Electronics covers / outdoor trim	ASA	UV resistance and near-ABS heat performance.	ABS indoors; SLA for visual and fit checks only.
Cycloidal discs / loaded gear teeth	Machined metal preferred	High-cycle contact stress and wear are poor matches for most printed polymers.	PA12-CF or PEEK only for controlled low-load experiments.

What not to use for primary structure

- Standard SLA resins for cyclic thigh, shank, chassis, or gearbox loads. Their clean surface and high tensile numbers do not remove UV aging, brittle fracture, micro-cracking, and poor impact tolerance.
- PLA-P for motor-adjacent or load-bearing robot parts. JLC3DP lists a 65 °C HDT and explicitly excludes prolonged heat and load-bearing use.
- Full-color PAC-HP or WJP resin for structural legs. Their purpose is appearance and detailed color, not high-cycle mechanics.
- As-printed holes or printed threads at motor mounts, bearings, pivots, or battery restraints. Use metal sleeves, through-bolts, inserts, or post-machining.

2. How to Read the Data

Property	Meaning	Why it matters on the robot
Tensile strength	Peak stress before tensile failure	Useful, but not a direct predictor of joint survival under bending, impact, fatigue, or layer separation.

Property	Meaning	Why it matters on the robot
Elastic / tensile modulus	Resistance to elastic deflection	Higher modulus makes links hold geometry better, but may reduce toughness and increase shock transmission.
Elongation at break	Ductility before fracture	A practical warning signal: PA11 at 50% is damage-tolerant; PA12-CF at 2.9% is much less forgiving.
Impact strength	Resistance to sudden energy input	Important for foot strikes, falls, backlash impacts, and joint hard stops.
HDT	Temperature at which a loaded specimen deflects	Always read the test load. PA11 is 185 °C at 0.45 MPa but only 54 °C at 1.82 MPa.
Process anisotropy	Direction-dependent properties	FDM Z-strength can be far below XY strength; powder-bed processes are generally more uniform.
Creep / fatigue	Time- and cycle-dependent deformation	Usually more important than one-time tensile strength for a walking robot; supplier tables rarely capture your exact duty cycle.

Comparison rule: Do not rank materials using one number from different test standards, orientations, moisture conditions, wall thicknesses, or infill settings. Use the published values for screening, then test production-intent coupons and a representative link or bracket.

3. Engineering Screening Scorecard

The heatmap converts the published data and process behavior into a quick 1-to-5 screening score. It is a design aid, not a supplier guarantee. Cost access includes process cost, likely finishing, and the risk of failed iterations.

Material family	Specific strength	Stiffness	Impact / fatigue	Heat	Process consistency	Cost access
MJF PA12-HP	4	3	4	4	5	4
MJF PA11-HP	4	3	5	4	5	3
SLS PA12 family	4	3	4	4	4	5
SLS 3401GB	3	4	2	4	4	4
FDM PA12-CF	4	5	2	4	2	4
FDM ASA / ABS	3	3	3	3	2	5
FDM PEEK	4	4	2	5	2	1
SLA engineering resin	2	3	1	2	5	5
TPU / PEBA	1	1	5	2	2	4
SLM TC4 titanium	5	5	4	5	4	1
SLM 316L	3	5	5	5	4	2
BJ 316L	3	5	4	5	3	3

Table 1. Relative quadruped-component screening score. Compare within a row and validate against the real load case.

Interpretation

- MJF PA12-HP is the safest general-purpose structural polymer; SLS PA12-family grades are close and often cost-effective.
- PA11-HP is the toughness leader; 3401GB and PA12-CF are stiffness-first choices with lower ductility.
- PEEK solves heat and chemical problems, not automatically impact or layer-bonding problems. JLC3DP publishes only 25.6 MPa tensile strength in Z versus 73 MPa in XY.
- Titanium offers the best printed-metal strength-to-weight ratio; 316L is heavier but more ductile and corrosion resistant.

4. Process-Level Comparison

Process	Accuracy / finish	Manufacturability	Main engineering limit	Cost	Robot fit
SLA resin	Very smooth; ±0.2 mm within 100 mm	Supports required; easy fine detail	UV/heat aging and brittle cyclic behavior	$–$$	Fit checks, molds, covers, visual prototypes
MJF nylon	Consistent, fine grain; ±0.3 mm	No support structures; good batch repeatability	Powder texture; moisture and creep still matter	$$	Primary body and leg structures
SLS nylon	Grainier; ±0.3 mm	No supports; complex hollow and ribbed geometry	Surface finish and dimensions less crisp than SLA	$–$$	Structural links, housings, prototypes
FDM plastic	Visible layers; ±0.3 mm	Large parts, familiar design rules	Strong anisotropy, warping, infill dependence	$–$$$	Covers, prototypes, stiffness-first PA12-CF parts
SLM metal	Rough; ±0.3 mm	Near-dense metal, complex topology	High cost, supports, stress relief and machining	$$$$	Compact high-load brackets and interfaces

Process	Accuracy / finish	Manufacturability	Main engineering limit	Cost	Robot fit
Binder-jet metal	More even than SLM; tolerance degrades >50 mm	No melt support strategy; economical batches	Sinter shrinkage, lower yield strength, 100 mm build limit	$$$	Small metal fittings, not precision joint geometry
WJP full color	Excellent color/detail; ±0.2 mm	Appearance-first workflow	Low heat/UV suitability and high cost	$$$	Team models, presentation and anatomy demonstrators

JLC3DP publishes general process tolerances of ±0.2 mm for SLA/WJP and ±0.3 mm for MJF, SLS, FDM, and SLM within 100 mm. Holes usually print undersize and require machining allowance.

5. Powder-Bed Nylon Materials

These are the most relevant materials for the robot's body and legs because they combine low density, toughness, complex geometry, and relatively uniform properties without FDM support structures.

Material	Process	Entry / relative cost	Tensile | modulus | elong.	Published HDT	Engineering judgment
PA12-HP	MJF	$1 / $$	48 MPa | 1.8 GPa | 20%	175 °C @0.45 MPa	Best all-round structural choice; strong, durable, chemically resistant.
PA11-HP	MJF	$1 / $$	52 MPa | 1.8 GPa | 50%	185 °C @0.45; 54 °C @1.82 MPa	Highest ductility and impact tolerance; excellent for clips and shock-prone links.
PAC-HP	MJF full color	$17 / $$$	No public strength table	80 °C	Color/detail material for models and craft prototypes, not a primary structure.
3301PA	SLS	$1 / $–$$	48 MPa | 1.7 GPa | 30%	147 °C @0.45; 55 °C @1.8 MPa	Tough white structural nylon; strong candidate for body and links.
3201PA-F	SLS	$1 / $–$$	44 MPa | 1.5 GPa | 35%	147 °C @0.45 MPa	Very tough, accurate, economical structural prototype and end-use option.
1172 Pro	SLS	$1 / $–$$	46 MPa | 1.8 GPa | 8–15%	179 °C @0.45 MPa	Good general mechanics and tall 590 mm build; lower ductility than PA12-HP/PA11.
3401GB	SLS glass-bead	$1 / $$	42 MPa | 2.5 GPa | 8%	150 °C @0.45; 60 °C @1.8 MPa	Stiffer and dimensionally stable; less impact tolerant. Use for housings/supports, not thin shock-loaded links.
PA12S-HP	MJF; support table	Quote / unknown	Dedicated public TDS not located	Verify current TDS	Appears in JLC3DP design tables; do not substitute for PA12-HP without the live material sheet.

Cost notation: the first value is JLC3DP's advertised starting price, not the final part quote. Relative tiers include process and finishing burden. JLC3DP's capability page contains a '3021PA-F' label while its material page and design guide use '3201PA-F'; verify the live quote name before ordering.

Nylon recommendation hierarchy

- Default: PA12-HP MJF for the main chassis and most leg shells.
- Toughness first: PA11-HP where falls, clips, flexing, or repeated shock dominate.
- Cost/large geometry: 3301PA or 3201PA-F SLS.
- Stiffness first: 3401GB for stable housings and supports, with generous radii and no sharp shock path.

6. FDM Plastics

Material	Entry / relative cost	Tensile | modulus | elong.	Thermal marker	Recommended role / limit
PLA-P	$1 / $	36–40 MPa | 2.55–2.68 GPa | 2.5–6%	65 °C	Cheap prototypes and jigs only; poor heat and long-term load behavior.
ABS	$3 / $	28.8 MPa | 1.85 GPa | 3.8%	98 °C	Indoor covers and thick simple functional parts; layer strength and warping limit structural use.
ASA	$3 / $$	43.8 MPa* | 2.38 GPa | 6.7%	97.8 °C	Best FDM outdoor cover material because of UV resistance. *Supplier page appears to mislabel the tensile unit; verify TDS.
PA12-CF	$3 / $$	69.3 MPa | 3.75 GPa | 2.9%	105 °C	Very stiff and strong in favorable orientation; low elongation and FDM anisotropy demand coupons and conservative joints.
TPU 95A	$3 / $$	No public tensile table on service page	Not listed	Replaceable foot pads, bumpers, cable strain relief; not a rigid structure.
PEBA 85A	$3 / $$$	21 MPa | 0.063 GPa | 600%	Low-temp capability to -70 °C	High elastic return for feet/lattices; less damping and harder process than TPU.
PEEK	$3 entry / $$$$ actual	XY 73, Z 25.6 MPa | ~3.0 GPa | 3.7/1.3%	144 °C @1.8 MPa	High-temperature/chemical specialty option; pronounced Z weakness and expensive process development.
ABS-ESD	$3 / $$	XY 38, Z 16 MPa | 2.0–2.5 GPa | 1.5–3.9%	Not listed	ESD-safe electronics fixtures and enclosures; not worth the strength penalty for ordinary robot structure.

Why PA12-CF is not automatically the winner

Its 3.75 GPa modulus is roughly twice PA12-HP's published 1.8 GPa, so it can reduce link flex. But its 2.9% elongation is far below PA12-HP's 20% and PA11-HP's 50%, and FDM layer interfaces remain orientation-dependent. Use it for a stiff shell with aligned load paths, not for thin snap features, hard-stop impacts, or an untested gearbox disc.

7. Printed Metals

Material	Process	Entry / relative cost	Published mechanics	Trade-off	Robot use
TC4 titanium	SLM	$8 / $$$$	1160 MPa tensile; 1000 MPa yield; 8% elong.	Best printed strength-to-weight option; rough and difficult to machine.	Weight-critical actuator brackets, pin carriers, compact high-load interfaces.
316L stainless	SLM	$8 / $$$	600 MPa tensile; 400 MPa yield; 30% elong.; E=180 GPa	Strong, ductile, corrosion resistant, but heavy; bores need post-machining.	Small bearing carriers, brackets, threaded bosses, safety-critical mounts.
BJ-316L	Binder jet	$5 / $$$	561 MPa tensile; 219 MPa yield; 50% elong.	Cheaper and smoother; shrink/warp risk and ±1.3% tolerance above 50 mm.	Small non-precision fittings and low-volume metal parts.

Weight warning: A steel bracket may solve a local stress problem while worsening whole-body inertia and leg torque demand. Prefer topology-optimized titanium only where the load path justifies the price; otherwise compare a machined aluminum part before choosing printed stainless steel.

8. SLA and Full-Color Resins

JLC3DP offers many resin names, but the structural differences are smaller than the gap between resin and powder-bed nylon. Treat most as precision prototype materials unless a dedicated fatigue and environmental test says otherwise.

Material	Entry price	Published differentiator	Best use / warning
Grey Resin	$0.30	HDT 51–59 °C; tensile 41–61 MPa; elong. 7–12%	Low-cost gray fit/appearance and light-duty prototype.
Black Resin	$1	HDT 58–69 °C; tensile 41–58 MPa; elong. 7–11%	Smooth black prototype; no UV/outdoor use.
JLC Black	$1	HDT 54–65 °C; supplier emphasizes hardness and finish	Batch appearance/functional prototypes, not cyclic legs.
Imagine Black	$2	HDT 46 °C; elong. 6–9%	High-precision/high-humidity appearance parts; low heat margin.
9600	$0.30	HDT 59 °C; matte white	Economical white enclosures, pipes, buckles, and fit checks.
9000HE	$0.30	HDT 60 °C; tensile 41–61 MPa; elong. 7–12%	General high-strength/tough resin prototype; still UV/heat limited.
8228	$0.30	HDT 56.4 °C; tensile 62.4 MPa; elong. 6.8%	Higher published tensile strength and stable batch prototypes.

Material	Entry price	Published differentiator	Best use / warning
8001	$0.30	Transparent/translucent; HDT 53 °C	Optical/flow/assembly visualization; default sanding and clear oil finish.
LEDO 6060	$1 on JLCPCB listing	HDT about 56 °C; dimensional stability and water-resistant prototype use	General housings and waterproof concept verification; not primary structure.
CBY	$0.30	HDT 72–80 °C; tensile 25–30 MPa; elong. 12–20%	More compliant/high-temperature resin for molds and functional prototypes.
JLC Temp	$0.30	HDT 101 °C; supplier warns it becomes brittle after cure	Static hot-air/tooling/wind-tunnel parts, not impacts or legs.
X Resin	$0.30	Randomly assigned resin/color	Cheap non-critical mockups only; impossible to engineer around an unknown grade.
Full-Color Resin	$5	WJP; HDT 60–65 °C; tensile 45–55 MPa; elong. 12–25%	Team communication models and presentation parts, not final robot structure.
Alternate names	Quote-dependent	JLCPCB pages also show 8000, 9000R, and 8111X	Treat as SLA prototype grades and verify the exact live TDS before substitution.

9. Design Rules for Robot Components

Design topic	Failure mechanism	Recommended practice
Wall thickness	Supplier minimum is a printability limit, not a structural recommendation.	Start structural nylon shells around 3–4 mm with ribs, then validate by FEA and test; thicken locally at joints.
Fasteners	Printed threads and point-loaded bosses crack or creep.	Use through-bolts, metal sleeves, washers, captured nuts, or M3–M5 inserts. Keep generous edge distance.

Design topic	Failure mechanism	Recommended practice
Bearings / shafts	As-printed bores shrink and wear.	Add machining allowance and use metal bearing cups or sleeves. Do not press a bearing directly into a thin resin wall.
Load paths	Sharp corners and sudden section changes concentrate stress.	Use large fillets, closed sections, ribs, and gradual transitions. Carry actuator load into multiple walls.
Powder removal	SLS/MJF powder remains trapped in closed cavities.	Add escape holes and inspection access; avoid inaccessible internal channels unless required.
FDM orientation	Z interfaces are the weakest direction.	Orient layer planes to carry bending and bolt loads; avoid peeling a flange across layers.
Thermal exposure	Motor and controller heat can reduce stiffness long before melting.	Measure local temperature under worst duty cycle and compare to HDT at the relevant stress, not only the 0.45 MPa number.
Replaceability	Feet, hard-stop pads, and cable guards are wear parts.	Make TPU/PEBA components replaceable without disassembling the whole leg.

10. Validation Plan Before Full-Robot Use

Step	Action	Acceptance intent
1	Freeze the load case	Record robot mass, peak actuator torque, joint geometry, hard-stop condition, fall height, motor temperature, and target life cycles.
2	Print coupons in production orientation	Test at least tensile/bend coupons plus a bolt-boss or bearing-seat coupon. Condition nylon in the expected humidity.
3	Test one representative component	Proof-load a thigh/shank or bracket gradually while measuring deflection. Inspect for white stress marks, cracks, bore growth, and insert movement.
4	Cycle under realistic motion	Run the component through repeated direction changes and impact-like loads. Static success does not establish fatigue life.
5	Instrument and inspect	Mark fasteners, measure temperatures, log joint error/current, and remeasure bores and critical dimensions after cycling.
6	Release with a documented margin	Keep a conservative safety factor for uncertainty and impacts; replace parts on measured wear or cycle count, not only after visible failure.

Recommended first procurement experiment

Order the same small structural bracket or scaled link section in PA12-HP, PA11-HP, 3301PA, 3401GB, and PA12-CF. Use identical holes and load fixtures. Compare mass, deflection, permanent set, impact damage, thread/insert retention, and final delivered price. That experiment will be more useful than selecting from tensile strength alone.

11. Final Component Selection

Robot component	Recommended material	Release condition
Body frame and battery/electronics tray	MJF PA12-HP	Ribbed shell, metal standoffs, through-bolts; keep heavy battery attachment in a distributed load path.
Thigh and shank	MJF PA12-HP or SLS 3301PA	Closed-section shell with large root fillets and metal pivot sleeves. Choose PA11 if falls and impacts dominate.

Robot component	Recommended material	Release condition
High-stiffness link revision	FDM PA12-CF	Only after orientation-specific coupon and full link proof test; avoid thin bosses and snap features.
Hip/actuator brackets	TC4 SLM for minimum mass; 316L SLM for lower material cost	Post-machine bearing and alignment surfaces. Compare CNC aluminum before committing.
Actuator/gearbox enclosure	PA12-CF shell plus metal bearing seats	Treat gear mesh, bearings, and threaded interfaces as metal/precision features.
Feet and impact pads	TPU 95A	Replaceable tread geometry; use PEBA only when rebound is intentionally desired.
Covers and guards	ASA outdoors; ABS indoors	Use SLA only when fine finish or visual verification outweighs impact and UV limits.

Bottom line: For the next weight-reduction body and leg build, choose PA12-HP MJF as the baseline and PA12-CF only as a targeted stiffness experiment. Use PA11-HP where cracking or impact is the failure mode. Keep the feet flexible and keep high-concentration bearing, shaft, and actuator loads in metal interfaces.

Sources and Data Boundaries

All service details below were checked on 26 August 2026. JLC3DP pages can change without notice. Mechanical values are reproduced or summarized from supplier pages and should be verified against the current quote-linked technical data sheet.

[1] JLC3DP 3D Printing Capabilities. Processes, material families, tolerances, finishing, and general design rules.

[2] JLC3DP 3D Printing Material Catalog. Current public catalog names for resin, nylon, plastic, and metal.

[3] JLC3DP Design Guideline. Build sizes, wall-thickness guidance, holes, and process tolerances.

[4] Functional 3D Printing Guide. Process behavior, functional-part boundaries, and validation cautions.

[5] PA12-HP Nylon. MJF PA12 price floor, dimensions, HDT, tensile, flexural, impact, and elongation.

[6] PA11-HP Nylon. MJF PA11 toughness, equal XY/Z data, elongation, and dual-load HDT.

[7] 3301PA Nylon. SLS 3301PA mechanical properties and limits.

[8] 3201PA-F Nylon. SLS 3201PA-F mechanical properties and print rules.

[9] Precimid 1172 Pro Nylon. SLS 1172 Pro mechanical properties and build envelope.

[10] 3401GB Nylon. Glass-bead nylon stiffness, strength, impact, elongation, and HDT.

[11] PA12-CF Plastic. FDM PA12-CF strength, stiffness, elongation, HDT, and service rules.

[12] PEEK Plastic. FDM PEEK XY/Z mechanics, thermal properties, and manufacturing limitations.

[13] ABS Plastic. FDM ABS mechanics, HDT, cost floor, build size, and default infill.

[14] ASA Plastic. FDM ASA UV use, mechanics, HDT, and service rules.

[15] TPU. FDM TPU hardness, applications, infill, and dimensions.

[16] PEBA Plastic. FDM PEBA elasticity, tensile strength, modulus, density, and low-temperature behavior.

[17] 316L Stainless Steel. SLM 316L strength, yield, elongation, modulus, roughness, and service rules.

[18] Titanium TC4. SLM TC4 strength, yield, elongation, and manufacturing limitations.

[19] BJ-316L Stainless Steel. Binder-jet 316L strength, yield, elongation, tolerance, and build envelope.

[20] JLC3DP Material Pricing Update. Published 2025 PA12 and SLM-316L price-per-gram changes; final quotes still depend on geometry and finishing.

Important limitations

- 'From' prices are marketing entry points, not normalized $/cm³ or final part prices. Shipping, minimum charge, geometry, packing, surface finish, color, batch volume, and special handling can dominate.
- Published strength may describe supplier test coupons rather than your exact infill, wall count, orientation, conditioning, or post-processing.
- HDT is not continuous-use temperature and is strongly load-dependent.
- Fatigue, creep, joint backlash, bearing-seat wear, and impact performance must be measured on representative robot parts.